

When Blocks Build Societies: Multi-Agent Systems in Minecraft

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Minecraft's open-world sandbox enables the study of complex multi-agent systems (MAS) in virtual environments

- Minecraft offers a procedurally generated, open-world environment with flexible mechanics
- Block-based game design balances computational feasibility with real-world complexity
- Supports a wide range of tasks like navigation, resource gathering, building, and social interaction

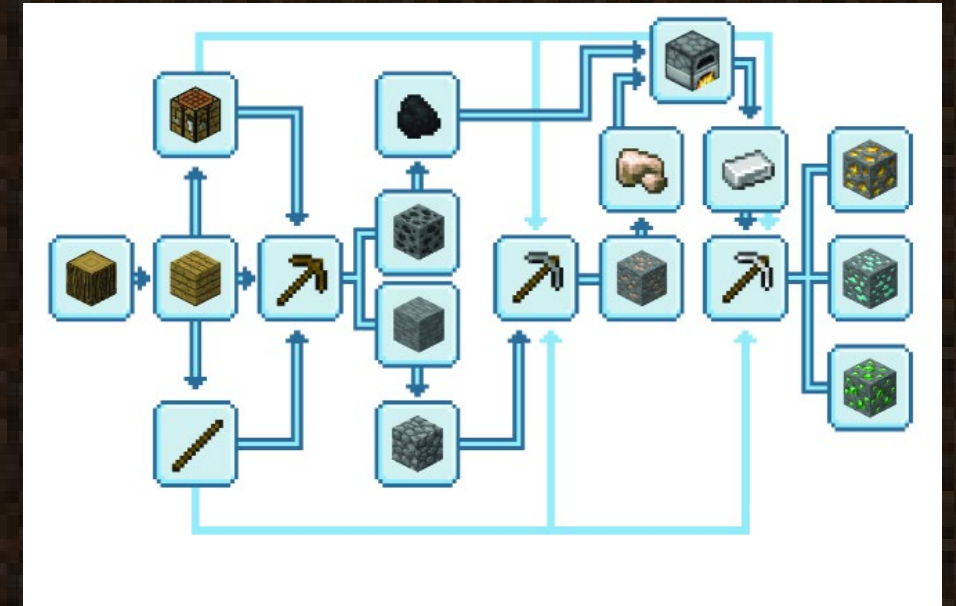


Figure 1: An example Minecraft technology dependency tree for the mining of gold, diamond, and emeralds.

How do different collaborative frameworks enable sustainable multi-agent collaboration in Minecraft?

- Identify design principles for robust, adaptable collaboration
- Explore theoretical and practical implications for future AI systems

Systematic comparative case study approach ensures transparent, replicable analysis of collaborative agent systems

- Selection criteria: relevance, documentation quality, and distinctiveness
- AI assisted with literature review and synthesis with human verification process and complete documentation for transparency
- Compare Project Sid (emergent/social) and TeamCraft (structured/task) along four dimensions:
 - Organization: How are MAS structured?
 - Communication: How do MAS coordinate?
 - Learning: How do MAS adapt?
 - Sustainability: How do MAS maintain long-term viability?

Project Sid shows that AI agents can develop social structures, roles, and cultures mirroring human societies

- Simulates 10–1000+ agents in Minecraft
- Emergent/social organization:
 - Agents develop specialized roles, informal rules, and cultural practices
 - Social structures and governance arise without explicit programming
- Features: self-organization, adaptability, and cultural transmission
- Limitations: initial inefficiency, computational constraints, and limited explicit control

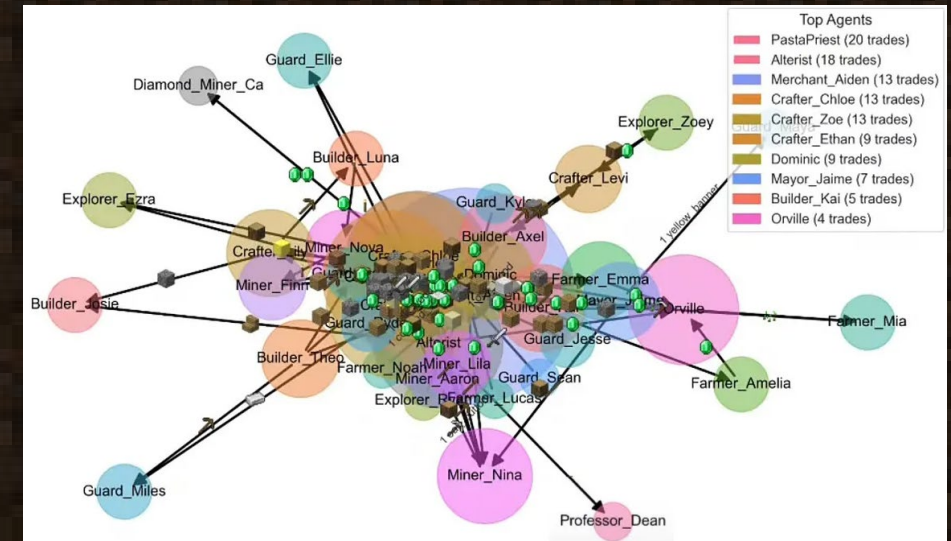


Figure 2: Trading activities among agents.

TeamCraft excels at evaluating and training agents for efficient multi-modal collaboration on structured tasks

- Features 55,000+ collaborative task in Minecraft
- Structured/task-oriented collaboration:
 - Explicit role assignment, task decomposition, and coordination protocols
 - Trains agents using visual/textual prompts and expert demonstrations
- Features: high efficiency on familiar tasks, clear protocols, measurable outcomes
- Limitations: struggles with generalization to novel tasks, limited long-term adaptability

System Prompt	Three Orthographic Views			
	Language Instruction	Three bots need to build a building on the platform. Write actions for <i>bot1</i> , <i>bot2</i> , <i>bot3</i> based on given observation .		
Observation	First Person View			
	Inventory Information	bot1 has 5 bricks, 3 iron_ore, bot2 has 2 sea_lantern, bot3 has 1 brick...		
Action		placeltem(bot1, 'bricks', (-1,0,-1)) placeltem(bot2, 'oak_planks', (0,0,0)) placeltem(bot3, 'iron_ore', (0,0,-1))		

Figure 3: Multi-modal prompt for building.

Comparing Project Sid with TeamCraft reveals trade-offs between generalization and efficiency...

Dimension	Project Sid	TeamCraft
Organization	Self-organizing, adaptive	Predefined, task-driven
Communication	Emergent, flexible	Structured, efficient
Learning	Reinforcement, social	Imitation learning
Adaptability	High for novel scenarios	High for known tasks, low for novel
Sustainability	Long-term resilience, scalable	Short-term efficiency, limited by training data

...while uncovering four design principles for collaborative systems that balance structure with flexibility

1. Combine structure and emergence for robust collaboration
2. Develop robust communication protocols
3. Support role flexibility and specialization
4. Enable cultural transmission for knowledge preservation

Insights from Minecraft can inform the design of adaptive, resilient AI for real-world collaboration

- Emergent organization (Project Sid) excels in dynamic, unpredictable environments where adaptability is key
- Structured coordination (TeamCraft) excels in stable, well-defined domains where efficiency is crucial
- Future likely lies in hybrid systems that can leverage both approaches

Next steps involve hybrid systems, long-term studies, and exploring transfer to real-world applications

- Develop and test hybrid frameworks combining structure and emergence
- Conduct long-term studies on evolution and stability of agent societies
- Investigate integration of MAS with human teams and transfer to physical world

References > When Blocks Build Societies: Multi-Agent Systems in Minecraft

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